

BIOLOGY — GRADE 10 | PLANT NUTRITION

Quick Check

Lesson 2: Leaf Structure: The Architecture of a Photosynthetic Machine

5 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.1: Plant Nutrition | Lesson 2

© 2026 SeaVuria and ARES. Aligned with KICD objectives. Developed by SeaVuria; technical implementation by ARES (with AI assistance from Claude). Licensed under CC BY-NC 4.0 (<https://creativecommons.org/licenses/by-nc/4.0/>).

Why is it useful that the cuticle is both waxy and transparent?

- A** It lets light in while reducing water loss
- B** It traps carbon dioxide inside the leaf
- C** It makes food for the leaf
- D** It carries water to the leaf cells

Which layer is packed tightly with chloroplasts to capture the most light?

- A** Lower epidermis
- B** Palisade mesophyll
- C** Spongy mesophyll
- D** Cuticle

What controls how wide a stoma opens?

- A** The cuticle
- B** The two guard cells around it
- C** The palisade cells
- D** The xylem vessels

Which tissue carries water from the roots up into the leaf?

A Phloem

B Xylem

C Palisade mesophyll

D Epidermis

Using today's evidence, why does the sukuma wiki wilt at midday even though the soil is moist?

- A** The soil dries out faster in strong sunlight
- B** Water vapour leaves through open stomata faster than xylem can replace it
- C** The roots stop absorbing water when it is hot
- D** Chlorophyll breaks down in direct sunlight

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).